Orange Pi 5 (RK3588S): which displays, resolutions and formats work

Mainline Linux 7.3 + this project's patches (the Ubuntu 26.10 image kernel). Every colour comes from hardware results on this board or from the hardware/kernel limits below; nothing is guessed. Data as of 2026-11-05.

How to read the chart (pages 2-7)

1. Find your resolution @ refresh row on the HDMI or DP pages. If your display uses a different timing, take its pixel clock and use the limits table on the right.
2. Pick the column for the colour format and bit depth you want (12 bit: not possible, the RK3588S HDMI and DP transmitters stop at 10 bit per colour; datasheet and driver max bpc 10).
3. The colour is the board's verdict; the text is the link it uses (TMDS rate, FRL rate x lanes, DP rate) or the limit that stops it, with the number. Then check that your display offers that mode and format (box below).

Colour key

VERIFIED	Works. Run on this board at exactly this timing, format and depth: accepted, wire format read back, camera saw a correct picture.
NOT TESTED	Not run on this board yet. Inside every hardware and driver limit, so it should work, but nobody has checked it on a real display (no display on the bench offers it, or no test covered it).
DRIVER LIMIT	Doesn't work yet: the RK3588S hardware can do it, the Linux driver can't (driver cap, missing feature or bug). The cell names the limit.
HARDWARE LIMIT	The RK3588S chip cannot output this uncompressed. The cell shows the number against the limit.
NOT DEFINED	The HDMI standard does not define this format at this timing, so no board and no display can use it. Not a board limit.

What your display must also support

The chart says what the board can send. The display decides what it accepts (its EDID):

- the mode itself (resolution, refresh, timing) must be in its list;
- HDMI 10 bit RGB / 4:4:4 needs Deep Color 30 in its HDMI VSDB; 4:2:2 needs 'YCbCr 4:2:2' in the CTA header;
- HDMI 4:2:0 only on modes it lists as 4:2:0 (Y420 VDB/CMDB), and 10 bit 4:2:0 needs DC_30 in its HF-VSDB;
- above 340 MHz TMDS it must support scrambling (HDMI 2.0); above 600 MHz, FRL with enough Gbit/s;
- DP: its maximum link rate and lane count (DPCD) must cover the cell; 4:2:0 needs VSC SDP support;
- HDR: PQ (ST 2084) or HLG in its HDR static metadata block, BT.2020 in its colorimetry block.

Our bench sinks and their limits: Appendix B. A red or orange cell will not work whatever the display supports.

Limits table: highest pixel clock (MHz) per link, format and depth

link	RGB/4:4:4 8b	RGB/4:4:4 10b	4:2:2 8/10b	4:2:0 8b	4:2:0 10b	status
HDMI TMDS <= 600 MHz	600	480	600	1200	960	verified
HDMI FRL 10G x4 = 40G (driver)	1481	1185	1481	2400	2370	verified
HDMI FRL 12G x4 = 48G (HW)	1778	1422	1778	2400	2400	driver cap
DP HBR2 x4 = 17.28G	720	576	1080/864	1440	1152	verified
DP HBR3 x4 = 25.92G	1080	864	1620/1296	2160	1728	not tested
DP HBR2 x2 = 8.64G (pin D)	360	288	540/432	720	576	not tested

Also for every row: VOP2 pixel clock <= 2400 MHz (hardware) and width <= 4096 px (driver; hardware VP0 7680 with splicing). Values capped at 2400 are VOP-limited. HDMI rule: TMDS if the character rate <= 600 MHz (rate = pixel clock x bpc/8; 4:2:2 = pixel clock; 4:2:0 half), otherwise FRL needing char rate x 27 bit/s. DP rule: pixel clock x bit/px <= lanes x rate x 0.8. DP YCbCr: currently sent as RGB (driver bug). Interlaced modes: not offered (interlace_allowed = 0 on both connectors).

Main findings

- HDMI: every TV mode/format combination the TV accepted works (346/346 camera GOOD, 64 refused by the TV's EDID). Up to 3840x2160@120 (FRL 10G x4) and 4096x2160@60, 8 and 10 bit, RGB/4:4:4/4:2:2/4:2:0.
- HDMI 4K120 10 bit RGB/4:4:4 needs 40.1 Gbit/s FRL: the driver's 10G/lane cap stops it; 4:2:2 10 bit works.
- DP: RGB 8/10 bit verified up to 3840x2160@60 on HBR2 x4 (370/370 GOOD). HBR3 untested (no HBR3 sink on the bench).
- DP YCbCr (4:4:4/4:2:2/4:2:0) is NOT output yet: the request is accepted and RGB is sent (driver bug, fix in progress).
- Above 4096 px wide (5120x1440, 8K): not supported by the driver (no VOP2 splice); 8K60 RGB also exceeds the HDMI link.
- Both outputs at once share one tunable video PLL: some HDMI FRL + DP pairs are refused (page 8).
- HDR10 (PQ, BT.2020, 10 bit) verified on both connectors, also one HDR and one SDR at once (page 9).

Pages

1 this overview, limits table, colour key
2 HDMI 720x400 ... 1280x1024
3 HDMI 1600x900 ... 3440x1440
4 HDMI 3440x1440 ... 7680x4320
5 DP 720x400 ... 1280x1024
6 DP 1600x900 ... 3440x1440
7 DP 3440x1440 ... 7680x4320
8 both outputs at once (shared video PLL)
9 HDR
10 Appendix A: limit codes, decision rules
11 Appendix B: bench displays, what was not tested

HDMI output (HDMI 2.1 TX, connector HDMI-A-1) [1/3]

1) find your resolution @ refresh row 2) pick the colour format + bit depth column 3) colour = can the board do it; text = the link it uses, or the limit that stops it. The display must also offer it (page 1).

Colour key: **VERIFIED** **NOT TESTED** **DRIVER LIMIT** **HARDWARE LIMIT** **NOT DEFINED** Non-green cells name the limit; Appendix A explains each

Resolution @ refresh (Hz) pixel clock MHz | timing | bench	RGB		YCbCr 4:4:4		YCbCr 4:2:2		YCbCr 4:2:0		Best working choice 10 bit first, then least subsampling
	8 bit 24 bit/px on wire	10 bit 30 bit/px on wire	8 bit 24 bit/px on wire	10 bit 30 bit/px on wire	8 bit 24 bit/px on wire	10 bit 24 bit/px on wire	8 bit 12 bit/px on wire	10 bit 15 bit/px on wire	
720x400 @70.08 28.320 | monitor EDID (DMT)	TMDS 28.3	TMDS 35.4	TMDS 28.3	TMDS 35.4	TMDS 28.3	TMDS 28.3	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 35.4 (not tested)
640x480 @60 / 59.94 TV 25.175 | CTA VIC 1	TMDS 25.2	8 bpc only VIC 1	TMDS 25.2	8 bpc only VIC 1	TMDS 25.2	8 bpc only VIC 1	TMDS 12.6<25 HDMI minimum	8 bpc only VIC 1	RGB 8b - TMDS 25.2
640x480 @66.67 30.240 | monitor EDID (DMT)	TMDS 30.2	TMDS 37.8	TMDS 30.2	TMDS 37.8	TMDS 30.2	TMDS 30.2	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 37.8 (not tested)
640x480 @72.81 31.500 | monitor EDID (DMT)	TMDS 31.5	TMDS 39.4	TMDS 31.5	TMDS 39.4	TMDS 31.5	TMDS 31.5	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 39.4 (not tested)
640x480 @75 31.500 | monitor EDID (DMT)	TMDS 31.5	TMDS 39.4	TMDS 31.5	TMDS 39.4	TMDS 31.5	TMDS 31.5	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 39.4 (not tested)
720x480 @60 / 59.94 TV 27.000 | CTA VIC 2	TMDS 27.0	TMDS 33.8	TMDS 27.0	TMDS 33.8	TMDS 27.0	TMDS 27.0	TMDS 13.5<25 HDMI minimum	TMDS 16.9<25 HDMI minimum	RGB 10b - TMDS 33.8
720x576 @50 27.000 | CTA VIC 17	TMDS 27.0	TMDS 33.8	TMDS 27.0	TMDS 33.8	TMDS 27.0	TMDS 27.0	TMDS 13.5<25 HDMI minimum	TMDS 16.9<25 HDMI minimum	RGB 10b - TMDS 33.8 (not tested)
800x600 @56.25 36.000 | monitor EDID (DMT)	TMDS 36.0	TMDS 45.0	TMDS 36.0	TMDS 45.0	TMDS 36.0	TMDS 36.0	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 45.0 (not tested)
800x600 @60.32 TV 40.000 | TV EDID	TMDS 40.0	TMDS 50.0	TMDS 40.0	TMDS 50.0	TMDS 40.0	TMDS 40.0	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 50.0
800x600 @72.19 50.000 | monitor EDID (DMT)	TMDS 50.0	TMDS 62.5	TMDS 50.0	TMDS 62.5	TMDS 50.0	TMDS 50.0	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 62.5 (not tested)
800x600 @75 49.500 | monitor EDID (DMT)	TMDS 49.5	TMDS 61.9	TMDS 49.5	TMDS 61.9	TMDS 49.5	TMDS 49.5	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 61.9 (not tested)
832x624 @74.55 57.284 | monitor EDID (DMT)	TMDS 57.3	TMDS 71.6	TMDS 57.3	TMDS 71.6	TMDS 57.3	TMDS 57.3	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 71.6 (not tested)
1024x768 @60 TV 65.000 | TV EDID	TMDS 65.0	TMDS 81.2	TMDS 65.0	TMDS 81.2	TMDS 65.0	TMDS 65.0	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 81.2
1024x768 @70.07 75.000 | monitor EDID (DMT)	TMDS 75.0	TMDS 93.8	TMDS 75.0	TMDS 93.8	TMDS 75.0	TMDS 75.0	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 93.8 (not tested)
1024x768 @75 78.750 | monitor EDID (DMT)	TMDS 78.8	TMDS 98.4	TMDS 78.8	TMDS 98.4	TMDS 78.8	TMDS 78.8	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 98.4 (not tested)
1280x720 @24 / 23.98 59.400 | CTA VIC 60	TMDS 59.4	TMDS 74.2	TMDS 59.4	TMDS 74.2	TMDS 59.4	TMDS 59.4	TMDS 29.7	TMDS 37.1	RGB 10b - TMDS 74.2 (not tested)
1280x720 @25 74.250 | CTA VIC 61	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 74.2	TMDS 37.1	TMDS 46.4	RGB 10b - TMDS 92.8 (not tested)
1280x720 @30 / 29.97 74.250 | CTA VIC 62	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 74.2	TMDS 37.1	TMDS 46.4	RGB 10b - TMDS 92.8 (not tested)
1280x720 @50 TV 74.250 | CTA VIC 19	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 74.2	TMDS 37.1 our TV refused	TMDS 46.4 our TV refused	RGB 10b - TMDS 92.8
1280x720 @60 / 59.94 TV 74.250 | CTA VIC 4	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 74.2	TMDS 37.1 our TV refused	TMDS 46.4 our TV refused	RGB 10b - TMDS 92.8
1280x720 @100 148.500 | CTA VIC 41	TMDS 148	TMDS 186	TMDS 148	TMDS 186	TMDS 148	TMDS 148	TMDS 74.2	TMDS 92.8	RGB 10b - TMDS 186 (not tested)
1280x720 @120 / 119.88 148.500 | CTA VIC 47	TMDS 148	TMDS 186	TMDS 148	TMDS 186	TMDS 148	TMDS 148	TMDS 74.2	TMDS 92.8	RGB 10b - TMDS 186 (not tested)
1152x864 @75 108.000 | monitor EDID (DMT)	TMDS 108	TMDS 135	TMDS 108	TMDS 135	TMDS 108	TMDS 108	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 135 (not tested)
1280x800 @59.81 83.500 | monitor EDID (DMT)	TMDS 83.5	TMDS 104	TMDS 83.5	TMDS 104	TMDS 83.5	TMDS 83.5	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 104 (not tested)
1366x768 @59.79 85.500 | VESA DMT	TMDS 85.5	TMDS 107	TMDS 85.5	TMDS 107	TMDS 85.5	TMDS 85.5	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 107 (not tested)
1280x960 @60 TV 108.000 | TV EDID	TMDS 108	TMDS 135	TMDS 108	TMDS 135	TMDS 108	TMDS 108	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 135
1440x900 @59.89 106.500 | monitor EDID (DMT)	TMDS 106	TMDS 133	TMDS 106	TMDS 133	TMDS 106	TMDS 106	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 133 (not tested)
1280x1024 @60 TV 108.000 | TV EDID	TMDS 108	TMDS 135	TMDS 108	TMDS 135	TMDS 108	TMDS 108	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 135
1280x1024 @75 135.000 | monitor EDID (DMT)	TMDS 135	TMDS 169	TMDS 135	TMDS 169	TMDS 135	TMDS 135	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 169 (not tested)

† FRL (HDMI 2.1 fixed-rate link): the driver always trains 10 Gbit/s x 4 lanes; about 2.7% of FRL mode sets leave the TV on 'No Signal' (4/150 trials, open bug). TMDS n = TMDS character rate in MHz (max 600). 'needs nG' = FRL bandwidth the mode needs (char rate x 27). 4:2:0 columns: only if the display lists that VIC as 4:2:0 (Y420); 4:2:2 is sent in a 12-bit container (24 bit/px) at any depth. 'our TV refused' = our TV's EDID did not allow it; the board can. Two rates (e.g. 60 / 59.94): the x1000/1001 twin behaves the same.

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1600x900 @60 TV 108.000 | TV EDID	TMDS 108	TMDS 135	TMDS 108	TMDS 135	TMDS 108	TMDS 108	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 135
1680x1050 @60 146.250 | monitor EDID (DMT)	TMDS 146	TMDS 183	TMDS 146	TMDS 183	TMDS 146	TMDS 146	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 183 (not tested)
1600x1200 @60 TV 162.000 | TV EDID	TMDS 162	TMDS 202	TMDS 162	TMDS 202	TMDS 162	TMDS 162	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 202
1600x1280 @60 TV 172.952 | TV EDID (GTF)	TMDS 173	TMDS 216	TMDS 173	TMDS 216	TMDS 173	TMDS 173	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 216
1920x1080 @24 / 23.98 TV 74.250 | CTA VIC 32	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 74.2	TMDS 37.1 our TV refused	TMDS 46.4 our TV refused	RGB 10b - TMDS 92.8
1920x1080 @25 74.250 | CTA VIC 33	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 74.2	TMDS 37.1	TMDS 46.4	RGB 10b - TMDS 92.8 (not tested)
1920x1080 @30 / 29.97 TV 74.250 | CTA VIC 34	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 92.8	TMDS 74.2	TMDS 74.2	TMDS 37.1 our TV refused	TMDS 46.4 our TV refused	RGB 10b - TMDS 92.8
1920x1080 @50 TV 148.500 | CTA VIC 31	TMDS 148	TMDS 186	TMDS 148	TMDS 186	TMDS 148	TMDS 148	TMDS 74.2 our TV refused	TMDS 92.8 our TV refused	RGB 10b - TMDS 186
1920x1080 @60 / 59.94 TV 148.500 | CTA VIC 16	TMDS 148	TMDS 186	TMDS 148	TMDS 186	TMDS 148	TMDS 148	TMDS 74.2 our TV refused	TMDS 92.8 our TV refused	RGB 10b - TMDS 186
1920x1080 @100 297.000 | CTA VIC 64	TMDS 297	TMDS 371	TMDS 297	TMDS 371	TMDS 297	TMDS 297	TMDS 148	TMDS 186	RGB 10b - TMDS 371 (not tested)
1920x1080 @120 / 119.88 TV 297.000 | CTA VIC 63	TMDS 297	TMDS 371	TMDS 297	TMDS 371	TMDS 297	TMDS 297	TMDS 148 our TV refused	TMDS 186 our TV refused	RGB 10b - TMDS 371
1920x1080 @144 333.216 | CVT-RB v2	TMDS 333	TMDS 417	TMDS 333	TMDS 417	TMDS 333	TMDS 333	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 417 (not tested)
1920x1080 @165 385.770 | CVT-RB v2	TMDS 386	TMDS 482	TMDS 386	TMDS 482	TMDS 386	TMDS 386	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 482 (not tested)
1920x1080 @240 583.200 | CVT-RB v2	TMDS 583	FRL 10G x4 † needs 19.7G	TMDS 583	FRL 10G x4 † needs 19.7G	TMDS 583	TMDS 583	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - FRL 10G x4 (not tested)
1920x1200 @60 154.000 | VESA DMT	TMDS 154	TMDS 192	TMDS 154	TMDS 192	TMDS 154	TMDS 154	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 192 (not tested)
1920x1200 @120 305.040 | CVT-RB v2	TMDS 305	TMDS 381	TMDS 305	TMDS 381	TMDS 305	TMDS 305	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 381 (not tested)
2560x1080 @60 175.982 | CVT-RB v2	TMDS 176	TMDS 220	TMDS 176	TMDS 220	TMDS 176	TMDS 176	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 220 (not tested)
2560x1080 @100 299.112 | CVT-RB v2	TMDS 299	TMDS 374	TMDS 299	TMDS 374	TMDS 299	TMDS 299	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 374 (not tested)
2560x1080 @144 439.845 | CVT-RB v2	TMDS 440	TMDS 550	TMDS 440	TMDS 550	TMDS 440	TMDS 440	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 550 (not tested)
2560x1440 @60 241.500 | monitor EDID (CVT-RB)	TMDS 242	TMDS 302	TMDS 242	TMDS 302	TMDS 242	TMDS 242	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 302 (not tested)
2560x1440 @75 295.416 | CVT-RB v2	TMDS 295	TMDS 369	TMDS 295	TMDS 369	TMDS 295	TMDS 295	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 369 (not tested)
2560x1440 @100 398.640 | CVT-RB v2	TMDS 399	TMDS 498	TMDS 399	TMDS 498	TMDS 399	TMDS 399	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 498 (not tested)
2560x1440 @120 TV 497.750 | TV EDID (DTD)	TMDS 498	FRL 10G x4 † needs 16.8G	TMDS 498	FRL 10G x4 † needs 16.8G	TMDS 498	TMDS 498	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - FRL 10G x4
2560x1440 @144 586.586 | CVT-RB v2	TMDS 587	FRL 10G x4 † needs 19.8G	TMDS 587	FRL 10G x4 † needs 19.8G	TMDS 587	TMDS 587	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - FRL 10G x4 (not tested)
2560x1440 @165 679.100 | CVT-RB v2	FRL 10G x4 † needs 18.3G	FRL 10G x4 † needs 22.9G	FRL 10G x4 † needs 18.3G	FRL 10G x4 † needs 22.9G	FRL 10G x4 † needs 18.3G	FRL 10G x4 † needs 18.3G	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - FRL 10G x4 (not tested)
2560x1440 @240 1025.798 | CVT-RB v2	FRL 10G x4 † needs 27.7G	FRL 10G x4 † needs 34.6G	FRL 10G x4 † needs 27.7G	FRL 10G x4 † needs 34.6G	FRL 10G x4 † needs 27.7G	FRL 10G x4 † needs 27.7G	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - FRL 10G x4 (not tested)
2560x1600 @60 268.500 | VESA DMT	TMDS 268	TMDS 336	TMDS 268	TMDS 336	TMDS 268	TMDS 268	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 336 (not tested)
2560x1600 @120 536.659 | CVT-RB v2	TMDS 537	FRL 10G x4 † needs 18.1G	TMDS 537	FRL 10G x4 † needs 18.1G	TMDS 537	TMDS 537	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - FRL 10G x4 (not tested)
3440x1440 @60 312.787 | CVT-RB v2	TMDS 313	TMDS 391	TMDS 313	TMDS 391	TMDS 313	TMDS 313	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - TMDS 391 (not tested)

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		8 bit 24 bit/px on wire	10 bit 30 bit/px on wire	8 bit 24 bit/px on wire	10 bit 30 bit/px on wire	8 bit 24 bit/px on wire	10 bit 24 bit/px on wire	8 bit 12 bit/px on wire	10 bit 15 bit/px on wire	
3440x1440 @100 531.520 | CVT-RB v2		TMDS 532	FRL 10G x4 † needs 17.9G	TMDS 532	FRL 10G x4 † needs 17.9G	TMDS 532	TMDS 532	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - FRL 10G x4 (not tested)
3440x1440 @144 782.115 | CVT-RB v2		FRL 10G x4 † needs 21.1G	FRL 10G x4 † needs 26.4G	FRL 10G x4 † needs 21.1G	FRL 10G x4 † needs 26.4G	FRL 10G x4 † needs 21.1G	FRL 10G x4 † needs 21.1G	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - FRL 10G x4 (not tested)
5120x1440 @60 462.072 | CVT-RB v2		5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	none: driver limit
5120x1440 @120 951.600 | CVT-RB v2		5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	none: driver limit
3840x2160 @24 / 23.98 297.000 | CTA VIC 93	TV	TMDS 297	TMDS 371	TMDS 297	TMDS 371	TMDS 297	TMDS 297	TMDS 148 our TV refused	TMDS 186 our TV refused	RGB 10b - TMDS 371
3840x2160 @25 297.000 | CTA VIC 94	TV	TMDS 297	TMDS 371	TMDS 297	TMDS 371	TMDS 297	TMDS 297	TMDS 148 our TV refused	TMDS 186 our TV refused	RGB 10b - TMDS 371
3840x2160 @30 / 29.97 297.000 | CTA VIC 95	TV	TMDS 297	TMDS 371	TMDS 297	TMDS 371	TMDS 297	TMDS 297	TMDS 148 our TV refused	TMDS 186 our TV refused	RGB 10b - TMDS 371
3840x2160 @50 594.000 | CTA VIC 96	TV	TMDS 594	FRL 10G x4 † needs 20.0G	TMDS 594	FRL 10G x4 † needs 20.0G	TMDS 594	TMDS 594	TMDS 297	TMDS 371	RGB 10b - FRL 10G x4
3840x2160 @60 533.250 | monitor EDID (CVT-RB)		TMDS 533	FRL 10G x4 † needs 18.0G	TMDS 533	FRL 10G x4 † needs 18.0G	TMDS 533	TMDS 533	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	RGB 10b - FRL 10G x4 (not tested)
3840x2160 @60 / 59.94 594.000 | CTA VIC 97	TV	TMDS 594	FRL 10G x4 † needs 20.0G	TMDS 594	FRL 10G x4 † needs 20.0G	TMDS 594	TMDS 594	TMDS 297	TMDS 371	RGB 10b - FRL 10G x4
3840x2160 @100 1188.000 | CTA VIC 117		FRL 10G x4 † needs 32.1G	40.1>40 Gb/s driver FRL cap	FRL 10G x4 † needs 32.1G	40.1>40 Gb/s driver FRL cap	FRL 10G x4 † needs 32.1G	FRL 10G x4 † needs 32.1G	TMDS 594	FRL 10G x4 † needs 20.0G	4:2:2 10b - FRL 10G x4 (not tested)
3840x2160 @120 / 119.88 1188.000 | CTA VIC 118	TV	FRL 10G x4 † needs 32.1G	40.1>40 Gb/s driver FRL cap	FRL 10G x4 † needs 32.1G	40.1>40 Gb/s driver FRL cap	FRL 10G x4 † needs 32.1G	FRL 10G x4 † needs 32.1G	TMDS 594	FRL 10G x4 † needs 20.0G	4:2:2 10b - FRL 10G x4
3840x2160 @144 1306.206 | CVT-RB v2		FRL 10G x4 † needs 35.3G	44.1>40 Gb/s driver FRL cap	FRL 10G x4 † needs 35.3G	44.1>40 Gb/s driver FRL cap	FRL 10G x4 † needs 35.3G	FRL 10G x4 † needs 35.3G	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	4:2:2 10b - FRL 10G x4 (not tested)
3840x2160 @165 1512.218 | CVT-RB v2		40.8>40 Gb/s driver FRL cap	51.0>48 Gb/s FRL max	40.8>40 Gb/s driver FRL cap	51.0>48 Gb/s FRL max	40.8>40 Gb/s driver FRL cap	40.8>40 Gb/s driver FRL cap	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	none: driver limit
3840x2160 @240 2285.203 | CVT-RB v2		61.7>48 Gb/s FRL max	77.1>48 Gb/s FRL max	61.7>48 Gb/s FRL max	77.1>48 Gb/s FRL max	61.7>48 Gb/s FRL max	61.7>48 Gb/s FRL max	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	none: hardware limit
4096x2160 @24 / 23.98 297.000 | CTA VIC 98	TV	TMDS 297	TMDS 371	TMDS 297	TMDS 371	TMDS 297	TMDS 297	TMDS 148 our TV refused	TMDS 186 our TV refused	RGB 10b - TMDS 371
4096x2160 @25 297.000 | CTA VIC 99		TMDS 297	TMDS 371	TMDS 297	TMDS 371	TMDS 297	TMDS 297	TMDS 148	TMDS 186	RGB 10b - TMDS 371 (not tested)
4096x2160 @30 / 29.97 297.000 | CTA VIC 100		TMDS 297	TMDS 371	TMDS 297	TMDS 371	TMDS 297	TMDS 297	TMDS 148	TMDS 186	RGB 10b - TMDS 371 (not tested)
4096x2160 @50 594.000 | CTA VIC 101	TV	TMDS 594	FRL 10G x4 † needs 20.0G	TMDS 594	FRL 10G x4 † needs 20.0G	TMDS 594	TMDS 594	TMDS 297	TMDS 371	RGB 10b - FRL 10G x4
4096x2160 @60 / 59.94 594.000 | CTA VIC 102	TV	TMDS 594	FRL 10G x4 † needs 20.0G	TMDS 594	FRL 10G x4 † needs 20.0G	TMDS 594	TMDS 594	TMDS 297	TMDS 371	RGB 10b - FRL 10G x4
4096x2160 @100 1188.000 | CTA VIC 218		FRL 10G x4 † needs 32.1G	40.1>40 Gb/s driver FRL cap	FRL 10G x4 † needs 32.1G	40.1>40 Gb/s driver FRL cap	FRL 10G x4 † needs 32.1G	FRL 10G x4 † needs 32.1G	TMDS 594	FRL 10G x4 † needs 20.0G	4:2:2 10b - FRL 10G x4 (not tested)
4096x2160 @120 / 119.88 1188.000 | CTA VIC 219		FRL 10G x4 † needs 32.1G	40.1>40 Gb/s driver FRL cap	FRL 10G x4 † needs 32.1G	40.1>40 Gb/s driver FRL cap	FRL 10G x4 † needs 32.1G	FRL 10G x4 † needs 32.1G	TMDS 594	FRL 10G x4 † needs 20.0G	4:2:2 10b - FRL 10G x4 (not tested)
5120x2880 @60 924.144 | CVT-RB v2		5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	5120>4096 px driver: 4096	no 4:2:0 no CTA VIC	no 4:2:0 no CTA VIC	none: driver limit
7680x4320 @24 / 23.98 1188.000 | CTA VIC 194		7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	none: driver limit
7680x4320 @25 1188.000 | CTA VIC 195		7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	none: driver limit
7680x4320 @30 / 29.97 1188.000 | CTA VIC 196		7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	7680>4096 px driver: 4096	none: driver limit
7680x4320 @50 2376.000 | CTA VIC 198		64.2>48 Gb/s FRL max	80.2>48 Gb/s FRL max	64.2>48 Gb/s FRL max	80.2>48 Gb/s FRL max	64.2>48 Gb/s FRL max	64.2>48 Gb/s FRL max	7680>4096 px driver: 4096	7680>4096 px driver: 4096	none: driver limit
7680x4320 @60 / 59.94 2376.000 | CTA VIC 199		64.2>48 Gb/s FRL max	80.2>48 Gb/s FRL max	64.2>48 Gb/s FRL max	80.2>48 Gb/s FRL max	64.2>48 Gb/s FRL max	64.2>48 Gb/s FRL max	7680>4096 px driver: 4096	7680>4096 px driver: 4096	none: driver limit
7680x4320 @120 / 119.88 4752.000 | CTA VIC 201		4752>2400MHz VOP max	4752>2400MHz VOP max	4752>2400MHz VOP max	4752>2400MHz VOP max	4752>2400MHz VOP max	4752>2400MHz VOP max	4752>2400MHz VOP max	4752>2400MHz VOP max	none: hardware limit

† FRL (HDMI 2.1 fixed-rate link): the driver always trains 10 Gbit/s x 4 lanes; about 2.7% of FRL mode sets leave the TV on 'No Signal' (4/150 trials, open bug). TMDS n = TMDS character rate in MHz (max 600). 'needs nG' = FRL bandwidth the mode needs (char rate x 27). 4:2:0 columns: only if the display lists that VIC as 4:2:0 (Y420); 4:2:2 is sent in a 12-bit container (24 bit/px) at any depth. 'our TV refused' = our TV's EDID did not allow it; the board can. Two rates (e.g. 60 / 59.94): the x1000/1001 twin behaves the same.

DisplayPort output (USB-C DP alt mode, 4 lanes, connector DP-1) [1/3]

1) find your resolution @ refresh row 2) pick the colour format + bit depth column 3) colour = can the board do it; text = the link it uses, or the limit that stops it. The display must also offer it (page 1).

Colour key: **VERIFIED** **NOT TESTED** **DRIVER LIMIT** **HARDWARE LIMIT** **NOT DEFINED** Non-green cells name the limit; Appendix A explains each

Resolution @ refresh (Hz) pixel clock MHz | timing | bench	RGB		YCbCr 4:4:4		YCbCr 4:2:2		YCbCr 4:2:0		Best working choice 10 bit first, then least subsampling
	8 bit 24 bit/px on wire	10 bit 30 bit/px on wire	8 bit 24 bit/px on wire	10 bit 30 bit/px on wire	8 bit 16 bit/px on wire	10 bit 20 bit/px on wire	8 bit 12 bit/px on wire	10 bit 15 bit/px on wire	
720x400 @70.08 monitor 28.320 | monitor EDID (DMT)	**0.7 Gb/s** RBR x4 13%	**0.8 Gb/s** RBR x4 16%	**tested: RGB** driver bug; needs 0.7G	**tested: RGB** driver bug; needs 0.8G	**tested: RGB** driver bug; needs 0.5G	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.3G	**tested: RGB** driver bug; needs 0.4G	**RGB 10b - RBR x4**
640x480 @60 / 59.94 25.175 | CTA VIC 1	**0.6 Gb/s** RBR x4 12%	**0.8 Gb/s** RBR x4 15%	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.8G	**tested: RGB** driver bug; needs 0.4G	**tested: RGB** driver bug; needs 0.5G	**tested: RGB** driver bug; needs 0.3G	**tested: RGB** driver bug; needs 0.4G	**RGB 10b - RBR x4**
640x480 @66.67 monitor 30.240 | monitor EDID (DMT)	**0.7 Gb/s** RBR x4 14%	**0.9 Gb/s** RBR x4 17%	**tested: RGB** driver bug; needs 0.7G	**tested: RGB** driver bug; needs 0.9G	**tested: RGB** driver bug; needs 0.5G	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.4G	**tested: RGB** driver bug; needs 0.5G	**RGB 10b - RBR x4**
640x480 @72.81 monitor 31.500 | monitor EDID (DMT)	**0.8 Gb/s** RBR x4 15%	**0.9 Gb/s** RBR x4 18%	**tested: RGB** driver bug; needs 0.8G	**tested: RGB** driver bug; needs 0.9G	**tested: RGB** driver bug; needs 0.5G	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.4G	**tested: RGB** driver bug; needs 0.5G	**RGB 10b - RBR x4**
640x480 @75 monitor 31.500 | monitor EDID (DMT)	**0.8 Gb/s** RBR x4 15%	**0.9 Gb/s** RBR x4 18%	**tested: RGB** driver bug; needs 0.8G	**tested: RGB** driver bug; needs 0.9G	**tested: RGB** driver bug; needs 0.5G	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.4G	**tested: RGB** driver bug; needs 0.5G	**RGB 10b - RBR x4**
720x480 @60 / 59.94 monitor 27.000 | CTA VIC 2	**0.6 Gb/s** RBR x4 12%	**0.8 Gb/s** RBR x4 16%	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.8G	**tested: RGB** driver bug; needs 0.4G	**tested: RGB** driver bug; needs 0.5G	**tested: RGB** driver bug; needs 0.3G	**tested: RGB** driver bug; needs 0.4G	**RGB 10b - RBR x4**
720x576 @50 monitor 27.000 | CTA VIC 17	**0.6 Gb/s** RBR x4 12%	**0.8 Gb/s** RBR x4 16%	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.8G	**tested: RGB** driver bug; needs 0.4G	**tested: RGB** driver bug; needs 0.5G	**tested: RGB** driver bug; needs 0.3G	**tested: RGB** driver bug; needs 0.4G	**RGB 10b - RBR x4**
800x600 @56.25 monitor 36.000 | monitor EDID (DMT)	**0.9 Gb/s** RBR x4 17%	**1.1 Gb/s** RBR x4 21%	**tested: RGB** driver bug; needs 0.9G	**tested: RGB** driver bug; needs 1.1G	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.7G	**tested: RGB** driver bug; needs 0.4G	**tested: RGB** driver bug; needs 0.5G	**RGB 10b - RBR x4**
800x600 @60.32 monitor 40.000 | TV EDID	**1.0 Gb/s** RBR x4 19%	**1.2 Gb/s** RBR x4 23%	**tested: RGB** driver bug; needs 1.0G	**tested: RGB** driver bug; needs 1.2G	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.8G	**tested: RGB** driver bug; needs 0.5G	**tested: RGB** driver bug; needs 0.6G	**RGB 10b - RBR x4**
800x600 @72.19 monitor 50.000 | monitor EDID (DMT)	**1.2 Gb/s** RBR x4 23%	**1.5 Gb/s** RBR x4 29%	**tested: RGB** driver bug; needs 1.2G	**tested: RGB** driver bug; needs 1.5G	**tested: RGB** driver bug; needs 0.8G	**tested: RGB** driver bug; needs 1.0G	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.8G	**RGB 10b - RBR x4**
800x600 @75 monitor 49.500 | monitor EDID (DMT)	**1.2 Gb/s** RBR x4 23%	**1.5 Gb/s** RBR x4 29%	**tested: RGB** driver bug; needs 1.2G	**tested: RGB** driver bug; needs 1.5G	**tested: RGB** driver bug; needs 0.8G	**tested: RGB** driver bug; needs 1.0G	**tested: RGB** driver bug; needs 0.6G	**tested: RGB** driver bug; needs 0.7G	**RGB 10b - RBR x4**
832x624 @74.55 monitor 57.284 | monitor EDID (DMT)	**1.4 Gb/s** RBR x4 27%	**1.7 Gb/s** RBR x4 33%	**tested: RGB** driver bug; needs 1.4G	**tested: RGB** driver bug; needs 1.7G	**tested: RGB** driver bug; needs 0.9G	**tested: RGB** driver bug; needs 1.1G	**tested: RGB** driver bug; needs 0.7G	**tested: RGB** driver bug; needs 0.9G	**RGB 10b - RBR x4**
1024x768 @60 monitor 65.000 | TV EDID	**1.6 Gb/s** RBR x4 30%	**1.9 Gb/s** RBR x4 38%	**tested: RGB** driver bug; needs 1.6G	**tested: RGB** driver bug; needs 1.9G	**tested: RGB** driver bug; needs 1.0G	**tested: RGB** driver bug; needs 1.3G	**tested: RGB** driver bug; needs 0.8G	**tested: RGB** driver bug; needs 1.0G	**RGB 10b - RBR x4**
1024x768 @70.07 monitor 75.000 | monitor EDID (DMT)	**1.8 Gb/s** RBR x4 35%	**2.2 Gb/s** RBR x4 43%	**tested: RGB** driver bug; needs 1.8G	**tested: RGB** driver bug; needs 2.2G	**tested: RGB** driver bug; needs 1.2G	**tested: RGB** driver bug; needs 1.5G	**tested: RGB** driver bug; needs 0.9G	**tested: RGB** driver bug; needs 1.1G	**RGB 10b - RBR x4**
1024x768 @75 monitor 78.750 | monitor EDID (DMT)	**1.9 Gb/s** RBR x4 36%	**2.4 Gb/s** RBR x4 46%	**tested: RGB** driver bug; needs 1.9G	**tested: RGB** driver bug; needs 2.4G	**tested: RGB** driver bug; needs 1.3G	**tested: RGB** driver bug; needs 1.6G	**tested: RGB** driver bug; needs 0.9G	**tested: RGB** driver bug; needs 1.2G	**RGB 10b - RBR x4**
1280x720 @24 / 23.98 59.400 | CTA VIC 60	**1.4 Gb/s** RBR x4 27%	**1.8 Gb/s** RBR x4 34%	**sends RGB** driver bug; needs 1.4G	**sends RGB** driver bug; needs 1.8G	**sends RGB** driver bug; needs 1.0G	**sends RGB** driver bug; needs 1.2G	**sends RGB** driver bug; needs 0.7G	**sends RGB** driver bug; needs 0.9G	**RGB 10b - RBR x4 (not tested)**
1280x720 @25 74.250 | CTA VIC 61	**1.8 Gb/s** RBR x4 34%	**2.2 Gb/s** RBR x4 43%	**sends RGB** driver bug; needs 1.8G	**sends RGB** driver bug; needs 2.2G	**sends RGB** driver bug; needs 1.2G	**sends RGB** driver bug; needs 1.5G	**sends RGB** driver bug; needs 0.9G	**sends RGB** driver bug; needs 1.1G	**RGB 10b - RBR x4 (not tested)**
1280x720 @30 / 29.97 74.250 | CTA VIC 62	**1.8 Gb/s** RBR x4 34%	**2.2 Gb/s** RBR x4 43%	**sends RGB** driver bug; needs 1.8G	**sends RGB** driver bug; needs 2.2G	**sends RGB** driver bug; needs 1.2G	**sends RGB** driver bug; needs 1.5G	**sends RGB** driver bug; needs 0.9G	**sends RGB** driver bug; needs 1.1G	**RGB 10b - RBR x4 (not tested)**
1280x720 @50 monitor 74.250 | CTA VIC 19	**1.8 Gb/s** RBR x4 34%	**2.2 Gb/s** RBR x4 43%	**tested: RGB** driver bug; needs 1.8G	**tested: RGB** driver bug; needs 2.2G	**tested: RGB** driver bug; needs 1.2G	**tested: RGB** driver bug; needs 1.5G	**tested: RGB** driver bug; needs 0.9G	**tested: RGB** driver bug; needs 1.1G	**RGB 10b - RBR x4**
1280x720 @60 / 59.94 monitor 74.250 | CTA VIC 4	**1.8 Gb/s** RBR x4 34%	**2.2 Gb/s** RBR x4 43%	**tested: RGB** driver bug; needs 1.8G	**tested: RGB** driver bug; needs 2.2G	**tested: RGB** driver bug; needs 1.2G	**tested: RGB** driver bug; needs 1.5G	**tested: RGB** driver bug; needs 0.9G	**tested: RGB** driver bug; needs 1.1G	**RGB 10b - RBR x4**
1280x720 @100 148.500 | CTA VIC 41	**3.6 Gb/s** RBR x4 69%	**4.5 Gb/s** RBR x4 86%	**sends RGB** driver bug; needs 3.6G	**sends RGB** driver bug; needs 4.5G	**sends RGB** driver bug; needs 2.4G	**sends RGB** driver bug; needs 3.0G	**sends RGB** driver bug; needs 1.8G	**sends RGB** driver bug; needs 2.2G	**RGB 10b - RBR x4 (not tested)**
1280x720 @120 / 119.88 148.500 | CTA VIC 47	**3.6 Gb/s** RBR x4 69%	**4.5 Gb/s** RBR x4 86%	**sends RGB** driver bug; needs 3.6G	**sends RGB** driver bug; needs 4.5G	**sends RGB** driver bug; needs 2.4G	**sends RGB** driver bug; needs 3.0G	**sends RGB** driver bug; needs 1.8G	**sends RGB** driver bug; needs 2.2G	**RGB 10b - RBR x4 (not tested)**
1152x864 @75 monitor 108.000 | monitor EDID (DMT)	**2.6 Gb/s** RBR x4 50%	**3.2 Gb/s** RBR x4 62%	**tested: RGB** driver bug; needs 2.6G	**tested: RGB** driver bug; needs 3.2G	**tested: RGB** driver bug; needs 1.7G	**tested: RGB** driver bug; needs 2.2G	**tested: RGB** driver bug; needs 1.3G	**tested: RGB** driver bug; needs 1.6G	**RGB 10b - RBR x4**
1280x800 @59.81 monitor 83.500 | monitor EDID (DMT)	**2.0 Gb/s** RBR x4 39%	**2.5 Gb/s** RBR x4 48%	**tested: RGB** driver bug; needs 2.0G	**tested: RGB** driver bug; needs 2.5G	**tested: RGB** driver bug; needs 1.3G	**tested: RGB** driver bug; needs 1.7G	**tested: RGB** driver bug; needs 1.0G	**tested: RGB** driver bug; needs 1.3G	**RGB 10b - RBR x4**
1366x768 @59.79 85.500 | VESA DMT	**2.1 Gb/s** RBR x4 40%	**2.6 Gb/s** RBR x4 49%	**sends RGB** driver bug; needs 2.1G	**sends RGB** driver bug; needs 2.6G	**sends RGB** driver bug; needs 1.4G	**sends RGB** driver bug; needs 1.7G	**sends RGB** driver bug; needs 1.0G	**sends RGB** driver bug; needs 1.3G	**RGB 10b - RBR x4 (not tested)**
1280x960 @60 108.000 | TV EDID	**2.6 Gb/s** RBR x4 50%	**3.2 Gb/s** RBR x4 62%	**sends RGB** driver bug; needs 2.6G	**sends RGB** driver bug; needs 3.2G	**sends RGB** driver bug; needs 1.7G	**sends RGB** driver bug; needs 2.2G	**sends RGB** driver bug; needs 1.3G	**sends RGB** driver bug; needs 1.6G	**RGB 10b - RBR x4 (not tested)**
1440x900 @59.89 monitor 106.500 | monitor EDID (DMT)	**2.6 Gb/s** RBR x4 49%	**3.2 Gb/s** RBR x4 62%	**tested: RGB** driver bug; needs 2.6G	**tested: RGB** driver bug; needs 3.2G	**tested: RGB** driver bug; needs 1.7G	**tested: RGB** driver bug; needs 2.1G	**tested: RGB** driver bug; needs 1.3G	**tested: RGB** driver bug; needs 1.6G	**RGB 10b - RBR x4**
1280x1024 @60 monitor 108.000 | TV EDID	**2.6 Gb/s** RBR x4 50%	**3.2 Gb/s** RBR x4 62%	**tested: RGB** driver bug; needs 2.6G	**tested: RGB** driver bug; needs 3.2G	**tested: RGB** driver bug; needs 1.7G	**tested: RGB** driver bug; needs 2.2G	**tested: RGB** driver bug; needs 1.3G	**tested: RGB** driver bug; needs 1.6G	**RGB 10b - RBR x4**
1280x1024 @75 monitor 135.000 | monitor EDID (DMT)	**3.2 Gb/s** RBR x4 62%	**4.0 Gb/s** RBR x4 78%	**tested: RGB** driver bug; needs 3.2G	**tested: RGB** driver bug; needs 4.0G	**tested: RGB** driver bug; needs 2.2G	**tested: RGB** driver bug; needs 2.7G	**tested: RGB** driver bug; needs 1.6G	**tested: RGB** driver bug; needs 2.0G	**RGB 10b - RBR x4**

Cell = data rate the mode needs (pixel clock x bit/px) and the smallest 4-lane DP link that carries it, with its use (payload after 8b/10b: RBR 5.18, HBR 8.64, HBR2 17.28, HBR3 25.92 Gbit/s). The driver trains the fastest rate both ends support. With only 2 lanes (USB-C pin D, USB 3 shared) the capacity halves. 'tested: RGB' = run on the monitor: accepted, picture GOOD, but the wire format was RGB. Highest DP pixel clock verified: 533.25 MHz; our monitor's sink limit is HBR2 x4 and 4K60.

DisplayPort output (USB-C DP alt mode, 4 lanes, connector DP-1) [2/3]

1) find your resolution @ refresh row 2) pick the colour format + bit depth column 3) colour = can the board do it; text = the link it uses, or the limit that stops it. The display must also offer it (page 1).

Colour key: **VERIFIED** **NOT TESTED** **DRIVER LIMIT** **HARDWARE LIMIT** **NOT DEFINED** Non-green cells name the limit; Appendix A explains each

Resolution @ refresh (Hz) pixel clock MHz | timing | bench	RGB		YCbCr 4:4:4		YCbCr 4:2:2		YCbCr 4:2:0		Best working choice 10 bit first, then least subsampling
	8 bit 24 bit/px on wire	10 bit 30 bit/px on wire	8 bit 24 bit/px on wire	10 bit 30 bit/px on wire	8 bit 16 bit/px on wire	10 bit 20 bit/px on wire	8 bit 12 bit/px on wire	10 bit 15 bit/px on wire	
1600x900 @60 monitor 108.000 | TV EDID	**2.6 Gb/s** RBR x4 50%	**3.2 Gb/s** RBR x4 62%	**tested: RGB** driver bug; needs 2.6G	**tested: RGB** driver bug; needs 3.2G	**tested: RGB** driver bug; needs 1.7G	**tested: RGB** driver bug; needs 2.2G	**tested: RGB** driver bug; needs 1.3G	**tested: RGB** driver bug; needs 1.6G	**RGB 10b - RBR x4**
1680x1050 @60 monitor 146.250 | monitor EDID (DMT)	**3.5 Gb/s** RBR x4 68%	**4.4 Gb/s** RBR x4 85%	**tested: RGB** driver bug; needs 3.5G	**tested: RGB** driver bug; needs 4.4G	**tested: RGB** driver bug; needs 2.3G	**tested: RGB** driver bug; needs 2.9G	**tested: RGB** driver bug; needs 1.8G	**tested: RGB** driver bug; needs 2.2G	**RGB 10b - RBR x4**
1600x1200 @60 162.000 | TV EDID	**3.9 Gb/s** RBR x4 75%	**4.9 Gb/s** RBR x4 94%	**sends RGB** driver bug; needs 3.9G	**sends RGB** driver bug; needs 4.9G	**sends RGB** driver bug; needs 2.6G	**sends RGB** driver bug; needs 3.2G	**sends RGB** driver bug; needs 1.9G	**sends RGB** driver bug; needs 2.4G	**RGB 10b - RBR x4 (not tested)**
1600x1280 @60 172.952 | TV EDID (GTF)	**4.2 Gb/s** RBR x4 80%	**5.2 Gb/s** HBR x4 60%	**sends RGB** driver bug; needs 4.2G	**sends RGB** driver bug; needs 5.2G	**sends RGB** driver bug; needs 2.8G	**sends RGB** driver bug; needs 3.5G	**sends RGB** driver bug; needs 2.1G	**sends RGB** driver bug; needs 2.6G	**RGB 10b - HBR x4 (not tested)**
1920x1080 @24 / 23.98 74.250 | CTA VIC 32	**1.8 Gb/s** RBR x4 34%	**2.2 Gb/s** RBR x4 43%	**sends RGB** driver bug; needs 1.8G	**sends RGB** driver bug; needs 2.2G	**sends RGB** driver bug; needs 1.2G	**sends RGB** driver bug; needs 1.5G	**sends RGB** driver bug; needs 0.9G	**sends RGB** driver bug; needs 1.1G	**RGB 10b - RBR x4 (not tested)**
1920x1080 @25 74.250 | CTA VIC 33	**1.8 Gb/s** RBR x4 34%	**2.2 Gb/s** RBR x4 43%	**sends RGB** driver bug; needs 1.8G	**sends RGB** driver bug; needs 2.2G	**sends RGB** driver bug; needs 1.2G	**sends RGB** driver bug; needs 1.5G	**sends RGB** driver bug; needs 0.9G	**sends RGB** driver bug; needs 1.1G	**RGB 10b - RBR x4 (not tested)**
1920x1080 @30 / 29.97 74.250 | CTA VIC 34	**1.8 Gb/s** RBR x4 34%	**2.2 Gb/s** RBR x4 43%	**sends RGB** driver bug; needs 1.8G	**sends RGB** driver bug; needs 2.2G	**sends RGB** driver bug; needs 1.2G	**sends RGB** driver bug; needs 1.5G	**sends RGB** driver bug; needs 0.9G	**sends RGB** driver bug; needs 1.1G	**RGB 10b - RBR x4 (not tested)**
1920x1080 @50 monitor 148.500 | CTA VIC 31	**3.6 Gb/s** RBR x4 69%	**4.5 Gb/s** RBR x4 86%	**tested: RGB** driver bug; needs 3.6G	**tested: RGB** driver bug; needs 4.5G	**tested: RGB** driver bug; needs 2.4G	**tested: RGB** driver bug; needs 3.0G	**tested: RGB** driver bug; needs 1.8G	**tested: RGB** driver bug; needs 2.2G	**RGB 10b - RBR x4**
1920x1080 @60 / 59.94 monitor 148.500 | CTA VIC 16	**3.6 Gb/s** RBR x4 69%	**4.5 Gb/s** RBR x4 86%	**tested: RGB** driver bug; needs 3.6G	**tested: RGB** driver bug; needs 4.5G	**tested: RGB** driver bug; needs 2.4G	**tested: RGB** driver bug; needs 3.0G	**tested: RGB** driver bug; needs 1.8G	**tested: RGB** driver bug; needs 2.2G	**RGB 10b - RBR x4**
1920x1080 @100 297.000 | CTA VIC 64	**7.1 Gb/s** HBR x4 82%	**8.9 Gb/s** HBR2 x4 52%	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**sends RGB** driver bug; needs 4.8G	**sends RGB** driver bug; needs 5.9G	**sends RGB** driver bug; needs 3.6G	**sends RGB** driver bug; needs 4.5G	**RGB 10b - HBR2 x4 (not tested)**
1920x1080 @120 / 119.88 297.000 | CTA VIC 63	**7.1 Gb/s** HBR x4 82%	**8.9 Gb/s** HBR2 x4 52%	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**sends RGB** driver bug; needs 4.8G	**sends RGB** driver bug; needs 5.9G	**sends RGB** driver bug; needs 3.6G	**sends RGB** driver bug; needs 4.5G	**RGB 10b - HBR2 x4 (not tested)**
1920x1080 @144 333.216 | CVT-RB v2	**8.0 Gb/s** HBR x4 93%	**10.0 Gb/s** HBR2 x4 58%	**sends RGB** driver bug; needs 8.0G	**sends RGB** driver bug; needs 10.0G	**sends RGB** driver bug; needs 5.3G	**sends RGB** driver bug; needs 6.7G	**sends RGB** driver bug; needs 4.0G	**sends RGB** driver bug; needs 5.0G	**RGB 10b - HBR2 x4 (not tested)**
1920x1080 @165 385.770 | CVT-RB v2	**9.3 Gb/s** HBR2 x4 54%	**11.6 Gb/s** HBR2 x4 67%	**sends RGB** driver bug; needs 9.3G	**sends RGB** driver bug; needs 11.6G	**sends RGB** driver bug; needs 6.2G	**sends RGB** driver bug; needs 7.7G	**sends RGB** driver bug; needs 4.6G	**sends RGB** driver bug; needs 5.8G	**RGB 10b - HBR2 x4 (not tested)**
1920x1080 @240 583.200 | CVT-RB v2	**14.0 Gb/s** HBR2 x4 81%	**17.5 Gb/s** HBR3 x4 67%	**sends RGB** driver bug; needs 14.0G	**sends RGB** driver bug; needs 17.5G	**sends RGB** driver bug; needs 9.3G	**sends RGB** driver bug; needs 11.7G	**sends RGB** driver bug; needs 7.0G	**sends RGB** driver bug; needs 8.7G	**RGB 10b - HBR3 x4 (not tested)**
1920x1200 @60 154.000 | VESA DMT	**3.7 Gb/s** RBR x4 71%	**4.6 Gb/s** RBR x4 89%	**sends RGB** driver bug; needs 3.7G	**sends RGB** driver bug; needs 4.6G	**sends RGB** driver bug; needs 2.5G	**sends RGB** driver bug; needs 3.1G	**sends RGB** driver bug; needs 1.8G	**sends RGB** driver bug; needs 2.3G	**RGB 10b - RBR x4 (not tested)**
1920x1200 @120 305.040 | CVT-RB v2	**7.3 Gb/s** HBR x4 85%	**9.2 Gb/s** HBR2 x4 53%	**sends RGB** driver bug; needs 7.3G	**sends RGB** driver bug; needs 9.2G	**sends RGB** driver bug; needs 4.9G	**sends RGB** driver bug; needs 6.1G	**sends RGB** driver bug; needs 3.7G	**sends RGB** driver bug; needs 4.6G	**RGB 10b - HBR2 x4 (not tested)**
2560x1080 @60 175.982 | CVT-RB v2	**4.2 Gb/s** RBR x4 81%	**5.3 Gb/s** HBR x4 61%	**sends RGB** driver bug; needs 4.2G	**sends RGB** driver bug; needs 5.3G	**sends RGB** driver bug; needs 2.8G	**sends RGB** driver bug; needs 3.5G	**sends RGB** driver bug; needs 2.1G	**sends RGB** driver bug; needs 2.6G	**RGB 10b - HBR x4 (not tested)**
2560x1080 @100 299.112 | CVT-RB v2	**7.2 Gb/s** HBR x4 83%	**9.0 Gb/s** HBR2 x4 52%	**sends RGB** driver bug; needs 7.2G	**sends RGB** driver bug; needs 9.0G	**sends RGB** driver bug; needs 4.8G	**sends RGB** driver bug; needs 6.0G	**sends RGB** driver bug; needs 3.6G	**sends RGB** driver bug; needs 4.5G	**RGB 10b - HBR2 x4 (not tested)**
2560x1080 @144 439.845 | CVT-RB v2	**10.6 Gb/s** HBR2 x4 61%	**13.2 Gb/s** HBR2 x4 76%	**sends RGB** driver bug; needs 10.6G	**sends RGB** driver bug; needs 13.2G	**sends RGB** driver bug; needs 7.0G	**sends RGB** driver bug; needs 8.8G	**sends RGB** driver bug; needs 5.3G	**sends RGB** driver bug; needs 6.6G	**RGB 10b - HBR2 x4 (not tested)**
2560x1440 @60 monitor 241.500 | monitor EDID (CVT-RB)	**5.8 Gb/s** HBR x4 67%	**7.2 Gb/s** HBR x4 84%	**tested: RGB** driver bug; needs 5.8G	**tested: RGB** driver bug; needs 7.2G	**tested: RGB** driver bug; needs 3.9G	**tested: RGB** driver bug; needs 4.8G	**tested: RGB** driver bug; needs 2.9G	**tested: RGB** driver bug; needs 3.6G	**RGB 10b - HBR x4**
2560x1440 @75 295.416 | CVT-RB v2	**7.1 Gb/s** HBR x4 82%	**8.9 Gb/s** HBR2 x4 51%	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**sends RGB** driver bug; needs 4.7G	**sends RGB** driver bug; needs 5.9G	**sends RGB** driver bug; needs 3.5G	**sends RGB** driver bug; needs 4.4G	**RGB 10b - HBR2 x4 (not tested)**
2560x1440 @100 398.640 | CVT-RB v2	**9.6 Gb/s** HBR2 x4 55%	**12.0 Gb/s** HBR2 x4 69%	**sends RGB** driver bug; needs 9.6G	**sends RGB** driver bug; needs 12.0G	**sends RGB** driver bug; needs 6.4G	**sends RGB** driver bug; needs 8.0G	**sends RGB** driver bug; needs 4.8G	**sends RGB** driver bug; needs 6.0G	**RGB 10b - HBR2 x4 (not tested)**
2560x1440 @120 497.750 | TV EDID (DTD)	**11.9 Gb/s** HBR2 x4 69%	**14.9 Gb/s** HBR2 x4 86%	**sends RGB** driver bug; needs 11.9G	**sends RGB** driver bug; needs 14.9G	**sends RGB** driver bug; needs 8.0G	**sends RGB** driver bug; needs 10.0G	**sends RGB** driver bug; needs 6.0G	**sends RGB** driver bug; needs 7.5G	**RGB 10b - HBR2 x4 (not tested)**
2560x1440 @144 586.586 | CVT-RB v2	**14.1 Gb/s** HBR2 x4 81%	**17.6 Gb/s** HBR3 x4 68%	**sends RGB** driver bug; needs 14.1G	**sends RGB** driver bug; needs 17.6G	**sends RGB** driver bug; needs 9.4G	**sends RGB** driver bug; needs 11.7G	**sends RGB** driver bug; needs 7.0G	**sends RGB** driver bug; needs 8.8G	**RGB 10b - HBR3 x4 (not tested)**
2560x1440 @165 679.100 | CVT-RB v2	**16.3 Gb/s** HBR2 x4 94%	**20.4 Gb/s** HBR3 x4 79%	**sends RGB** driver bug; needs 16.3G	**sends RGB** driver bug; needs 20.4G	**sends RGB** driver bug; needs 10.9G	**sends RGB** driver bug; needs 13.6G	**sends RGB** driver bug; needs 8.1G	**sends RGB** driver bug; needs 10.2G	**RGB 10b - HBR3 x4 (not tested)**
2560x1440 @240 1025.798 | CVT-RB v2	**24.6 Gb/s** HBR3 x4 95%	**30.8>25.9G** > HBR3 x4	**sends RGB** driver bug; needs 24.6G	**30.8>25.9G** > HBR3 x4	**sends RGB** driver bug; needs 16.4G	**sends RGB** driver bug; needs 20.5G	**sends RGB** driver bug; needs 12.3G	**sends RGB** driver bug; needs 15.4G	**RGB 8b - HBR3 x4 (not tested)**
2560x1600 @60 268.500 | VESA DMT	**6.4 Gb/s** HBR x4 75%	**8.1 Gb/s** HBR x4 93%	**sends RGB** driver bug; needs 6.4G	**sends RGB** driver bug; needs 8.1G	**sends RGB** driver bug; needs 4.3G	**sends RGB** driver bug; needs 5.4G	**sends RGB** driver bug; needs 3.2G	**sends RGB** driver bug; needs 4.0G	**RGB 10b - HBR x4 (not tested)**
2560x1600 @120 536.659 | CVT-RB v2	**12.9 Gb/s** HBR2 x4 75%	**16.1 Gb/s** HBR2 x4 93%	**sends RGB** driver bug; needs 12.9G	**sends RGB** driver bug; needs 16.1G	**sends RGB** driver bug; needs 8.6G	**sends RGB** driver bug; needs 10.7G	**sends RGB** driver bug; needs 6.4G	**sends RGB** driver bug; needs 8.0G	**RGB 10b - HBR2 x4 (not tested)**
3440x1440 @60 312.787 | CVT-RB v2	**7.5 Gb/s** HBR x4 87%	**9.4 Gb/s** HBR2 x4 54%	**sends RGB** driver bug; needs 7.5G	**sends RGB** driver bug; needs 9.4G	**sends RGB** driver bug; needs 5.0G	**sends RGB** driver bug; needs 6.3G	**sends RGB** driver bug; needs 3.8G	**sends RGB** driver bug; needs 4.7G	**RGB 10b - HBR2 x4 (not tested)**

Cell = data rate the mode needs (pixel clock x bit/px) and the smallest 4-lane DP link that carries it, with its use (payload after 8b/10b: RBR 5.18, HBR 8.64, HBR2 17.28, HBR3 25.92 Gbit/s). The driver trains the fastest rate both ends support. With only 2 lanes (USB-C pin D, USB 3 shared) the capacity halves. 'tested: RGB' = run on the monitor: accepted, picture GOOD, but the wire format was RGB. Highest DP pixel clock verified: 533.25 MHz; our monitor's sink limit is HBR2 x4 and 4K60.

DisplayPort output (USB-C DP alt mode, 4 lanes, connector DP-1) [3/3]

1) find your resolution @ refresh row 2) pick the colour format + bit depth column 3) colour = can the board do it; text = the link it uses, or the limit that stops it. The display must also offer it (page 1).

Colour key: **VERIFIED** **NOT TESTED** **DRIVER LIMIT** **HARDWARE LIMIT** **NOT DEFINED** Non-green cells name the limit; Appendix A explains each

Resolution @ refresh (Hz) pixel clock MHz | timing | bench	RGB		YCbCr 4:4:4		YCbCr 4:2:2		YCbCr 4:2:0		Best working choice 10 bit first, then least subsampling
	8 bit 24 bit/px on wire	10 bit 30 bit/px on wire	8 bit 24 bit/px on wire	10 bit 30 bit/px on wire	8 bit 16 bit/px on wire	10 bit 20 bit/px on wire	8 bit 12 bit/px on wire	10 bit 15 bit/px on wire	
3440x1440 @100 531.520 | CVT-RB v2	**12.8 Gb/s** HBR2 x4 74%	**15.9 Gb/s** HBR2 x4 92%	**sends RGB** driver bug; needs 12.8G	**sends RGB** driver bug; needs 15.9G	**sends RGB** driver bug; needs 8.5G	**sends RGB** driver bug; needs 10.6G	**sends RGB** driver bug; needs 6.4G	**sends RGB** driver bug; needs 8.0G	**RGB 10b - HBR2 x4 (not tested)**
3440x1440 @144 782.115 | CVT-RB v2	**18.8 Gb/s** HBR3 x4 72%	**23.5 Gb/s** HBR3 x4 91%	**sends RGB** driver bug; needs 18.8G	**sends RGB** driver bug; needs 23.5G	**sends RGB** driver bug; needs 12.5G	**sends RGB** driver bug; needs 15.6G	**sends RGB** driver bug; needs 9.4G	**sends RGB** driver bug; needs 11.7G	**RGB 10b - HBR3 x4 (not tested)**
5120x1440 @60 462.072 | CVT-RB v2	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**none: driver limit**
5120x1440 @120 951.600 | CVT-RB v2	**5120>4096 px** driver: 4096	**28.5>25.9G** > HBR3 x4	**5120>4096 px** driver: 4096	**28.5>25.9G** > HBR3 x4	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**none: driver limit**
3840x2160 @24 / 23.98 297.000 | CTA VIC 93	**7.1 Gb/s** HBR x4 82%	**8.9 Gb/s** HBR2 x4 52%	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**sends RGB** driver bug; needs 4.8G	**sends RGB** driver bug; needs 5.9G	**sends RGB** driver bug; needs 3.6G	**sends RGB** driver bug; needs 4.5G	**RGB 10b - HBR2 x4 (not tested)**
3840x2160 @25 297.000 | CTA VIC 94	**7.1 Gb/s** HBR x4 82%	**8.9 Gb/s** HBR2 x4 52%	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**sends RGB** driver bug; needs 4.8G	**sends RGB** driver bug; needs 5.9G	**sends RGB** driver bug; needs 3.6G	**sends RGB** driver bug; needs 4.5G	**RGB 10b - HBR2 x4 (not tested)**
3840x2160 @30 / 29.97 monitor 297.000 | CTA VIC 95	**7.1 Gb/s** HBR x4 82%	**8.9 Gb/s** HBR2 x4 52%	**tested: RGB** driver bug; needs 7.1G	**tested: RGB** driver bug; needs 8.9G	**tested: RGB** driver bug; needs 4.8G	**tested: RGB** driver bug; needs 5.9G	**tested: RGB** driver bug; needs 3.6G	**tested: RGB** driver bug; needs 4.5G	**RGB 10b - HBR2 x4**
3840x2160 @50 594.000 | CTA VIC 96	**14.3 Gb/s** HBR2 x4 82%	**17.8 Gb/s** HBR3 x4 69%	**sends RGB** driver bug; needs 14.3G	**sends RGB** driver bug; needs 17.8G	**sends RGB** driver bug; needs 9.5G	**sends RGB** driver bug; needs 11.9G	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**RGB 10b - HBR3 x4 (not tested)**
3840x2160 @60 monitor 533.250 | monitor EDID (CVT-RB)	**12.8 Gb/s** HBR2 x4 74%	**16.0 Gb/s** HBR2 x4 93%	**tested: RGB** driver bug; needs 12.8G	**tested: RGB** driver bug; needs 16.0G	**tested: RGB** driver bug; needs 8.5G	**tested: RGB** driver bug; needs 10.7G	**tested: RGB** driver bug; needs 6.4G	**tested: RGB** driver bug; needs 8.0G	**RGB 10b - HBR2 x4**
3840x2160 @60 / 59.94 594.000 | CTA VIC 97	**14.3 Gb/s** HBR2 x4 82%	**17.8 Gb/s** HBR3 x4 69%	**sends RGB** driver bug; needs 14.3G	**sends RGB** driver bug; needs 17.8G	**sends RGB** driver bug; needs 9.5G	**sends RGB** driver bug; needs 11.9G	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**RGB 10b - HBR3 x4 (not tested)**
3840x2160 @100 1188.000 | CTA VIC 117	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**19.0 Gb/s** HBR3 x4 73%	**23.8 Gb/s** HBR3 x4 92%	**sends RGB** driver bug; needs 14.3G	**sends RGB** driver bug; needs 17.8G	**4:2:2 10b - HBR3 x4 (not tested)**
3840x2160 @120 / 119.88 1188.000 | CTA VIC 118	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**19.0 Gb/s** HBR3 x4 73%	**23.8 Gb/s** HBR3 x4 92%	**sends RGB** driver bug; needs 14.3G	**sends RGB** driver bug; needs 17.8G	**4:2:2 10b - HBR3 x4 (not tested)**
3840x2160 @144 1306.206 | CVT-RB v2	**31.3>25.9G** > HBR3 x4	**39.2>25.9G** > HBR3 x4	**31.3>25.9G** > HBR3 x4	**39.2>25.9G** > HBR3 x4	**20.9 Gb/s** HBR3 x4 81%	**26.1>25.9G** > HBR3 x4	**sends RGB** driver bug; needs 15.7G	**sends RGB** driver bug; needs 19.6G	**4:2:2 8b - HBR3 x4 (not tested)**
3840x2160 @165 1512.218 | CVT-RB v2	**36.3>25.9G** > HBR3 x4	**45.4>25.9G** > HBR3 x4	**36.3>25.9G** > HBR3 x4	**45.4>25.9G** > HBR3 x4	**24.2 Gb/s** HBR3 x4 93%	**30.2>25.9G** > HBR3 x4	**sends RGB** driver bug; needs 18.1G	**sends RGB** driver bug; needs 22.7G	**4:2:2 8b - HBR3 x4 (not tested)**
3840x2160 @240 2285.203 | CVT-RB v2	**54.8>25.9G** > HBR3 x4	**68.6>25.9G** > HBR3 x4	**54.8>25.9G** > HBR3 x4	**68.6>25.9G** > HBR3 x4	**36.6>25.9G** > HBR3 x4	**45.7>25.9G** > HBR3 x4	**27.4>25.9G** > HBR3 x4	**34.3>25.9G** > HBR3 x4	**none: hardware limit**
4096x2160 @24 / 23.98 297.000 | CTA VIC 98	**7.1 Gb/s** HBR x4 82%	**8.9 Gb/s** HBR2 x4 52%	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**sends RGB** driver bug; needs 4.8G	**sends RGB** driver bug; needs 5.9G	**sends RGB** driver bug; needs 3.6G	**sends RGB** driver bug; needs 4.5G	**RGB 10b - HBR2 x4 (not tested)**
4096x2160 @25 297.000 | CTA VIC 99	**7.1 Gb/s** HBR x4 82%	**8.9 Gb/s** HBR2 x4 52%	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**sends RGB** driver bug; needs 4.8G	**sends RGB** driver bug; needs 5.9G	**sends RGB** driver bug; needs 3.6G	**sends RGB** driver bug; needs 4.5G	**RGB 10b - HBR2 x4 (not tested)**
4096x2160 @30 / 29.97 297.000 | CTA VIC 100	**7.1 Gb/s** HBR x4 82%	**8.9 Gb/s** HBR2 x4 52%	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**sends RGB** driver bug; needs 4.8G	**sends RGB** driver bug; needs 5.9G	**sends RGB** driver bug; needs 3.6G	**sends RGB** driver bug; needs 4.5G	**RGB 10b - HBR2 x4 (not tested)**
4096x2160 @50 594.000 | CTA VIC 101	**14.3 Gb/s** HBR2 x4 82%	**17.8 Gb/s** HBR3 x4 69%	**sends RGB** driver bug; needs 14.3G	**sends RGB** driver bug; needs 17.8G	**sends RGB** driver bug; needs 9.5G	**sends RGB** driver bug; needs 11.9G	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**RGB 10b - HBR3 x4 (not tested)**
4096x2160 @60 / 59.94 594.000 | CTA VIC 102	**14.3 Gb/s** HBR2 x4 82%	**17.8 Gb/s** HBR3 x4 69%	**sends RGB** driver bug; needs 14.3G	**sends RGB** driver bug; needs 17.8G	**sends RGB** driver bug; needs 9.5G	**sends RGB** driver bug; needs 11.9G	**sends RGB** driver bug; needs 7.1G	**sends RGB** driver bug; needs 8.9G	**RGB 10b - HBR3 x4 (not tested)**
4096x2160 @100 1188.000 | CTA VIC 218	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**19.0 Gb/s** HBR3 x4 73%	**23.8 Gb/s** HBR3 x4 92%	**sends RGB** driver bug; needs 14.3G	**sends RGB** driver bug; needs 17.8G	**4:2:2 10b - HBR3 x4 (not tested)**
4096x2160 @120 / 119.88 1188.000 | CTA VIC 219	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**19.0 Gb/s** HBR3 x4 73%	**23.8 Gb/s** HBR3 x4 92%	**sends RGB** driver bug; needs 14.3G	**sends RGB** driver bug; needs 17.8G	**4:2:2 10b - HBR3 x4 (not tested)**
5120x2880 @60 924.144 | CVT-RB v2	**5120>4096 px** driver: 4096	**27.7>25.9G** > HBR3 x4	**5120>4096 px** driver: 4096	**27.7>25.9G** > HBR3 x4	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**5120>4096 px** driver: 4096	**none: driver limit**
7680x4320 @24 / 23.98 1188.000 | CTA VIC 194	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**7680>4096 px** driver: 4096	**7680>4096 px** driver: 4096	**7680>4096 px** driver: 4096	**7680>4096 px** driver: 4096	**none: driver limit**
7680x4320 @25 1188.000 | CTA VIC 195	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**7680>4096 px** driver: 4096	**7680>4096 px** driver: 4096	**7680>4096 px** driver: 4096	**7680>4096 px** driver: 4096	**none: driver limit**
7680x4320 @30 / 29.97 1188.000 | CTA VIC 196	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**7680>4096 px** driver: 4096	**7680>4096 px** driver: 4096	**7680>4096 px** driver: 4096	**7680>4096 px** driver: 4096	**none: driver limit**
7680x4320 @50 2376.000 | CTA VIC 198	**57.0>25.9G** > HBR3 x4	**71.3>25.9G** > HBR3 x4	**57.0>25.9G** > HBR3 x4	**71.3>25.9G** > HBR3 x4	**38.0>25.9G** > HBR3 x4	**47.5>25.9G** > HBR3 x4	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**none: hardware limit**
7680x4320 @60 / 59.94 2376.000 | CTA VIC 199	**57.0>25.9G** > HBR3 x4	**71.3>25.9G** > HBR3 x4	**57.0>25.9G** > HBR3 x4	**71.3>25.9G** > HBR3 x4	**38.0>25.9G** > HBR3 x4	**47.5>25.9G** > HBR3 x4	**28.5>25.9G** > HBR3 x4	**35.6>25.9G** > HBR3 x4	**none: hardware limit**
7680x4320 @120 / 119.88 4752.000 | CTA VIC 201	**4752>2400MHz** VOP max	**4752>2400MHz** VOP max	**4752>2400MHz** VOP max	**4752>2400MHz** VOP max	**4752>2400MHz** VOP max	**4752>2400MHz** VOP max	**4752>2400MHz** VOP max	**4752>2400MHz** VOP max	**none: hardware limit**

Cell = data rate the mode needs (pixel clock x bit/px) and the smallest 4-lane DP link that carries it, with its use (payload after 8b/10b: RBR 5.18, HBR 8.64, HBR2 17.28, HBR3 25.92 Gbit/s). The driver trains the fastest rate both ends support. With only 2 lanes (USB-C pin D, USB 3 shared) the capacity halves. 'tested: RGB' = run on the monitor: accepted, picture GOOD, but the wire format was RGB. Highest DP pixel clock verified: 533.25 MHz; our monitor's sink limit is HBR2 x4 and 4K60.

Both outputs at once (HDMI + DP): the shared video PLL

HDMI drives video port VP0, DP drives VP2. Each port's pixel clock (DCLK) must come from somewhere; one tunable PLL (V0PLL) is shared by both. Model = the kernel's VOP2 PLL planner, checked against hardware.

Where the pixel clock comes from:
• HDMI in TMDS mode (character rate <= 600 MHz): the HDMI PHY's own PLL, any rate. Never clashes.
• HDMI in FRL mode: an EXACT integer divide (VP0 divider <= 128) of GPLL 1188 MHz, CPLL 1500 MHz or AUPLL 786.432 MHz (fixed, never retuned), otherwise V0PLL tuned to a multiple of it. HDMI must be exact.
• DP: a divide (VP2 divider <= 32) of GPLL/CPLL/AUPLL within 0.5% of the mode (VESA tolerance), otherwise V0PLL.
• Only when BOTH need V0PLL can they clash: they work if one V0PLL rate gives HDMI exactly and DP within 0.5%. Otherwise the kernel refuses the second mode set with EINVAL (no silent off-rate clock). This is a hardware limit: the RK3588S has one free display PLL for two ports, and the HDMI PHY PLL cannot clock a second port during FRL (measured: no pixels). Fix on the user side: pick HDMI 4:2:2 10 bit (stays TMDS) or another DP mode.

Grid: HDMI FRL configurations that need V0PLL (rows: HDMI pixel clock, modes, formats) x DP modes that need V0PLL (columns: DP pixel clock + mode). Every pair NOT in this grid works (they never share a PLL).

DP pixel clock MHz, mode →	25.175 640x480@60	25.200 640x480@60	28.320 720x400@70	36.000 800x600@56	40.000 800x600@60	57.284 832x624@75	85.500 1366x768@60	106.500 1440x900@60	135.000 1280x1024@75	146.250 1680x1050@60	154.000 1920x1200@60	162.000 1600x1200@60	172.952 1600x1280@60	175.982 2560x1080@60	241.500 2560x1440@60	268.500 2560x1600@60	295.416 2560x1440@75	305.040 1920x1200@120	312.787 3440x1440@60	333.216 1920x1080@144	385.770 1920x1080@165	398.640 2560x1440@100	439.845 2560x1080@144	531.520 3440x1440@100	533.250 3840x2160@60	536.659 2560x1600@120	583.200 1920x1080@240	586.586 2560x1440@144	679.100 2560x1440@165	782.115 3440x1440@144	1025.798 2560x1440@240
497.750 MHz 2560x1440@120 FRL: 4:4:4 10b, RGB 10b	✗	✗	✗	✗	ok	ok	✗	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	ok	✗	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗
531.520 MHz 3440x1440@100 FRL: 4:4:4 10b, RGB 10b	✗	ok	✗	✗	✗	✗	✗	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗	ok	✗	✗	✗	ok	✗	ok	ok	✗	✗	✗	✗	✗	✗
533.250 MHz 3840x2160@60 FRL: 4:4:4 10b, RGB 10b	✗	✗	✗	✗	✗	ok	ok	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗	ok	✗	✗	✗	ok	✗	ok	ok	✗	✗	✗	✗	✗	✗
536.659 MHz 2560x1600@120 FRL: 4:4:4 10b, RGB 10b	✗	✗	ok	✗	✗	ok	ok	✗	✗	ok	ok	✗	✗	✗	✗	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗	ok	✗	✗	✗	✗	✗
583.200 MHz 1920x1080@240 FRL: 4:4:4 10b, RGB 10b	✗	✗	✗	✗	✗	✗	✗	ok	ok	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗	ok	✗	✗	✗	✗	✗	✗	ok	✗	✗	✗	✗
586.586 MHz 2560x1440@144 FRL: 4:4:4 10b, RGB 10b	✗	✗	✗	✗	✗	✗	✗	ok	ok	ok	✗	✗	✗	ok	✗	✗	✗	✗	✗	✗	✗	✗	ok	✗	✗	✗	✗	ok	✗	ok	✗
593.407 MHz 3840/4096x2160@60, 3840/4096x2160@120 FRL: 4:2:0 10b, 4:4:4 10b, RGB 10b	✗	✗	ok	✗	✗	ok	✗	✗	✗	✗	✗	ok	✗	✗	✗	✗	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
679.100 MHz 2560x1440@165 FRL: 4:2:2 10b, 4:2:2 8b, 4:4:4 10b, 4:4:4 8b, RGB 10b, RGB 8b	ok	ok	ok	✗	ok	✗	✗	✗	✗	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	ok	✗	✗
782.115 MHz 3440x1440@144 FRL: 4:2:2 10b, 4:2:2 8b, 4:4:4 10b, 4:4:4 8b, RGB 10b, RGB 8b	ok	ok	✗	✗	✗	✗	✗	ok	✗	ok	✗	✗	ok	✗	✗	✗	✗	✗	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗	ok	✗	ok	✗
1025.798 MHz 2560x1440@240 FRL: 4:2:2 10b, 4:2:2 8b, 4:4:4 10b, 4:4:4 8b, RGB 10b, RGB 8b	✗	✗	✗	✗	✗	✗	ok	✗	✗	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	ok
1186.813 MHz 3840/4096x2160@120 FRL: 4:2:2 10b, 4:2:2 8b, 4:4:4 8b, RGB 8b	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	ok	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
1306.206 MHz 3840x2160@144 FRL: 4:2:2 10b, 4:2:2 8b, 4:4:4 8b, RGB 8b	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗

works, pair run on hardware | should work (planner model, not run) | refused, confirmed on hardware (EINVAL) | refused by the model (hardware limit)

Hardware evidence (both heads on, camera on both screens): 275 pair commits in order + 368 in shuffled order, and 173 more in the regression run. Refused pairs: the same 7 every time, in both orders and from all-off, exactly the pairs the model refuses (also in the grid: ✗ on red). The model verdicts match all 816 hardware pair commits; over the full bench matrix (1549 pairs) it refuses only those 7. DP picture GOOD in every accepted pair. HDMI: 4 BLACK screens in the first 643 commits, all on FRL configurations (consistent with the FRL 'No Signal' bug, page 2; not yet re-tested), and in the regression run one pair (3840x2160@59.94 10 bit FRL + DP 1280x800@59.81) black 2/2, since traced to the same FRL 'No Signal' bug.
Other dual-head limits: both heads at 3840x2160@60 (and HDR on one, SDR on the other) verified 10/10. Memory fetch: each of the 2 VOP2 AXI read ports carries about one full-screen 4K@120 RGB layer (~4.4 GB/s); the kernel refuses plane setups that would overload a port (userspace then composites). 4:2:0 HDMI DCLK = pixel clock / 2. The V0PLL rate and DP clock error of every 'ok' pair are in dualhead.csv.

HDR (high dynamic range) output

HDR10 = PQ (SMPTE ST 2084) + BT.2020 + static metadata, 10 bit. Camera-verified on the bench TV (HDMI) and monitor (native DP).

feature	HDMI	DP	evidence / limit
HDR10: PQ EOTF + static metadata	verified	verified	HDMI DRM InfoFrame EOTF 2, max DML 1000; DP HDR metadata SDP; camera n75 SDR 0.86 / PQ 1.11 (TV), 0.70 / 1.53 (monitor)
BT.2020 colorimetry	verified	verified	HDMI AVI C=BT.2020; DP VSC SDP colorimetry (sink must support VSC SDP)
10 bit output with HDR	verified	verified	RGB 10 bit: HDMI 1080p60 TMDS, 2160p60 FRL; DP 1080p60/1440p60/2160p30/2160p60 (HBR2 x4)
HDR with 8 bit output	not tested	verified	legal, more banding (DP VSC SDP announces 8 bpc). DP: a '4:2:0 asked' run went out as RGB 8 bit + PQ, HDR PASS (table below)
HLG EOTF	not tested	not tested	accepted by the kernel (TEST_ONLY); not camera-checked. The kernel does not check the EDID: userspace must
SDR -> HDR -> SDR switching	verified	verified	metadata and colorimetry cleared on return; only the toggled head blanks briefly (modeset)
one head HDR, other SDR	verified	verified	16/16 steps at 1080p, 10/10 at 2160p60, unchanged head never disturbed
HDR with YCbCr 4:2:0/4:2:2 (HDMI)	not tested	driver bug	HDMI: BT.2020 YCC matrix in VOP2; DP: YCbCr not output yet (driver bug)
BT.2020 constant luminance (cYCC)	hardware limit	hardware limit	VOP2 has no constant-luminance encoder: refused in atomic check
Dynamic metadata (HDR10+, Dolby Vision)	no kernel API	no kernel API	mainline DRM has no UAPI for dynamic HDR metadata
Tone mapping / HDR engine in VOP2	driver doesn't use it	driver doesn't use it	the VP0 HDR10/SDR2HDR engine exists in hardware; mainline does not use it (pass-through only)

Hardware runs (2026-11-05)

connector	mode	wire format	link	SDR n75	PQ n75	back n75	verdict
HDMI (TV)	1920x1080@60	RGB 10 bit	TMDS 185.6	0.859	1.110	0.859	HDR PASS
HDMI (TV)	3840x2160@60	RGB 10 bit	FRL 10G x4	0.858	1.106	0.858	HDR PASS
DP (monitor)	1920x1080@60	RGB 10 bit	HBR2 x4	0.708	1.529	0.706	HDR PASS
DP (monitor)	2560x1440@60	RGB 10 bit	HBR2 x4	0.691	1.332	0.693	HDR PASS
DP (monitor)	3840x2160@30	RGB 10 bit	HBR2 x4	0.706	1.532	0.707	HDR PASS
DP (monitor)	3840x2160@60 (533.25)	RGB 10 bit	HBR2 x4 (16.0G)	0.706	1.544	0.705	HDR PASS
DP (monitor)	3840x2160@60, '4:2:0' asked	RGB 8 bit (bug)	HBR2 x4	0.700	1.542	0.700	driver bug: sent RGB

n75 = camera brightness of the 75% grey bar relative to white (lighting-invariant): SDR 0.7-0.9, HDR >= 1.0 because the display tone-maps PQ 75% (~1000 nit) to its peak. Each HDR case is judged against its own SDR control and 'back' case. Which modes can carry HDR: every green or grey 10 bit cell on pages 2-7 (HDR needs nothing beyond 10 bit + BT.2020 + metadata); our TV also lists HLG, our monitor PQ only.

Appendix A: the limits and how each status is decided

Every non-green cell is stopped by one of these limits. Full numbers per cell (pixel clock, bit/px, TMDS/FRL or DP rate, limit) are in cells.csv (one row per cell).

HDMI link over 48 Gb/s	Uncompressed signal needs more than FRL 12G x 4 lanes = 48 Gbit/s, the RK3588S HDMI 2.1 TX maximum (datasheet: FRL 3/6/8/10/12 Gbps). Need = TMDS character rate x 27 (3 ch x 8 bit x 18/16). DSC (in silicon) is not supported by mainline and not modelled.
VOP pixel clock over 2400 MHz	Pixel clock above the VOP2 video-port maximum 2400 MHz (rk3588 max_pixel_clock_rate 600 MHz x 4; VP0 tops out at 7680x4320@60 = 2376 MHz).
DP link over HBR3 x4	Needs more than 4 lanes x HBR3 8.1 Gbit/s x 8/10 = 25.92 Gbit/s, the DP 1.4a TX maximum. The RK3588S DP TX has no DSC.
driver caps FRL at 10G/lane	Fits FRL 12G x 4 = 48 Gbit/s (hardware OK) but the driver caps FRL at 10 Gbit/s per lane = 40 Gbit/s (rk3588_hdmi_cfg.max_frl_rate_per_lane = 10, upstream FIXME 'screen flickering above 10 Gbps/lane'). With 'max bpc' 10 the kernel silently drops to 8 bpc instead (verified at 4K120).
driver: VOP2 width max 4096	Wider than 4096 pixels: mainline VOP2 limits every RK3588 video port to 4096 (max_output.width). Hardware: VP0 alone can drive up to 7680 by splicing VP0+VP1 (vendor kernel), not in mainline; DP would also need re-routing from VP2 (max 4096 in hardware) to VP0.
DP driver bug: YCbCr request ignored	The driver accepts the DP 'color format' = YCbCr request but sends RGB (VOP2 bus format RGB888/RGB101010 on every DP YCbCr test, 222/222 cases; a 4:2:0 request was even accepted on a monitor without 4:2:0). Likely cause (code reading, not yet proven): drm_atomic_bridge_chain_select_bus_fmts() filters only the last bridge's formats, and the bridge after dw-dp reports MEDIA_BUS_FMT_FIXED, so dw-dp's own list is never filtered. Not a hardware limit (datasheet: DP TX RGB/YUV).
HDMI 4:2:0 only on CTA VICs	HDMI sends YCbCr 4:2:0 only for CTA-861 VIC timings that the display lists in its Y420 Video/Capability Map Data Block. This timing (VESA/CVT/EDID DTD) has no VIC, so no HDMI display can accept 4:2:0 here. Not a board limit.
VIC 1 is 8 bpc only	640x480 (VIC 1) is defined for 8 bpc only (drm_hdmi_state_helper follows the HDMI spec). A 10 bpc request gives 8 bpc (verified).
below HDMI minimum TMDS	TMDS character rate below 25 MHz, the HDMI minimum (4:2:0 halves the rate of an already slow mode). No HDMI display accepts it.
needs HBR3 (untested)	Fits only HBR3 x4 (25.92 Gbit/s). The driver trains up to HBR3 (dw-dp max_link_rate 810000, USBDP PHY 8100) but no HBR3 sink was on the bench (the monitor's DPCD max is HBR2).
DP pixel clock above tested	Within every limit, but above the highest DP pixel clock verified on hardware (533.25 MHz, 3840x2160@60 CVT-RB).
DP YCbCr only as fallback	The driver ignores the colour-format request on DP (bug), but its automatic choice falls back to this format here because RGB does not fit the link. Never seen on hardware.
HDMI path not yet run	Within every limit, but this format/depth was never run on this link type (TMDS or FRL) or the pixel clock is above the highest verified (1188 MHz).

Decision order per cell

HDMI: (1) not defined by the standard (4:2:0 without a CTA VIC; VIC 1 above 8 bpc; TMDS below 25 MHz) -> not defined (2) pixel clock > 2400 MHz or FRL need > 48 Gbit/s -> hardware limit (3) width > 4096 or FRL need > 40 Gbit/s -> driver limit (4) this exact cell ran correctly on this board -> verified; otherwise not tested.
DP: (1) pixel clock > 2400 MHz or need > HBR3 x4 25.92 Gbit/s -> hardware limit (2) width > 4096, or a YCbCr request (sent as RGB, bug) -> driver limit (3) this exact cell ran correctly -> verified; otherwise not tested.
Timings: CTA-861 VIC timings where a VIC exists; VESA DMT; modes from the bench EDIDs as listed; other rows use VESA CVT reduced blanking v2 (hblank 80, vblank >= 460 us). Another timing for the same resolution changes the pixel clock: then use the limits table on page 1.

Appendix B: the bench displays (their limits are not the board's)

Green results on pages 2-7 were run on these two displays. A mode or format they lack is not a board failure; 'our TV refused' marks cells our TV's EDID did not allow.

	TV: 'Beyond TV' (TCL, HDMI 2.1)	Monitor: Samsung LU28R55 (native DP over USB-C cable)
link	max TMDS 600 MHz, SCDC, FRL up to 12G x 4 lanes	DPCD 1.4, max HBR2 (5.4G), 4 lanes (pin C), VSC SDP colorimetry
max mode	3840x2160@120 (VIC 118), 2560x1440@120 DTD	3840x2160@60 CVT-RB 533.25 MHz, range 40-60 Hz, max 600 MHz
formats	RGB, YCbCr 4:4:4 / 4:2:2; 4:2:0 only on VIC 96, 97, 101, 102, 118	RGB, YCbCr 4:4:4 / 4:2:2 (no 4:2:0)
bit depth	Deep Color 30/36 RGB + 4:4:4; 4:2:0 DC 30/36/48	10 bit per colour (EDID)
HDR	PQ, HLG, BT.2020 RGB/YCC; Dolby Vision + HDR10+ blocks	PQ, BT.2020 RGB/YCC
modes listed	26 chart rows (41 modes incl. x1000/1001 twins)	29 chart rows (37 modes)
format test	410 cases: 346 camera GOOD, 64 refused by EDID rules	370 cases: 370 camera GOOD (YCbCr requests sent as RGB: driver bug)

What the board was NOT tested with

HBR3 DP sinks (an HBR3 adapter arrives 2026-11-05: HBR3 cells then become testable); DP 2-lane (pin D) setups on native DP; displays above 4K120; HDMI FRL rates other than 10G x 4 (the driver always picks the highest allowed); 4:2:0 on HDMI modes our TV does not list as 4:2:0 (e.g. 1080p 4:2:0); DP 4:2:0 (monitor lacks it, and the YCbCr bug).
Not shown as rows: interlaced modes (both connectors report interlace_allowed = 0, so the kernel never offers them); DSC (in the HDMI TX silicon, unsupported in mainline); 6 bpc DP.